BCS-5A CA Assignment # 2. 24K-0756.

Q1) (i) single cycle = 15 ns/inst.
pipelined ideal CPI = 1, so
3 ns/instr
speedup = $\frac{15}{3} = \boxed{5x}$

(ii) $CPI = 1 + 0.25 \times 1 = 1.25$
$time = 1.25 \times 3 = 3.75$ ns
speedup = $\frac{15}{3.75} = \boxed{4x}$

(iii) 4-stage => CPI = 1, t = 3.1 ns./inst.
5-stage => ~~CPI=~~ t = 3 ns/inst.
$\frac{3.1}{3} = 1.033$ → 4 stage is slower about 3.3%.

Same CPI, fewer stages but a longer cycle, so time per inst. is worse

(Q2) M1: resolves in ID, 1 delay slot, 80% filled. Penalty only when slot unfilled: $0.2 \times 1 = 0.2$ per branch.

$CPI = 1 + 0.35 \times 0.2 = 1.07$.

M2: resolves on EX predict not taken. Taken branches lose 2 cycles, not taken lose 0: $0.25 \times 2 = 0.5$ per branch
$CPI_2 = 1 + 0.35 \times 0.5 = 1.175$

Same cycle time, M1 is faster. Speedup = $\frac{1.175}{1.07} = \boxed{1.098x}$
about 9.8%

(Q3)(i) RAW: $I_1 \to I_7$ (R_2) ; $I_3 \to I_4$ (R_5)
$I_2 \to I_4$ (R_4) ; $I_4 \to I_5$ (R_6)
WAR: $I_2 \to I_6$ (R_3) ~~I~~
$I_3 \to I_6$ (R_3)
$I_4 \to I_5$ (R_4)
WAW: $I_2 \to I_5$ (R_4).

(ii) No forwarding, same cycle write + read: consumer ID must be in the producer's WB cycle. $I_3 \to I_4$ (R_5). I_3 WB = cyc. 7, I_4 ID moves from 5 to 7 → 2 stalls ($I_2 \to I_4$ is covered by the same wait)
$I_4 \to I_5$ (R_6): I_4 WB = cycle 10, I_5 must wait 2 more cycles → 2 stalls.
Branch I_7 is ~~unresolved~~ resolved in ID by ~~stall~~ stalling → 1 stall
No stall: $I_1 \to I_7$ (far apart). WAR & WAW never stall (in-order, read in ID, write in WB).
Cycles = $(8+4) + 2 + 2 + 1 = \boxed{17}$

(iii) Full forwarding + predict not taken. Loop runs once so branch is not taken, prediction is right, no penalty.
Still stall: $I_3 \to I_4$ only. I_3 is a load, R_5 is available only after MEM but I_4 needs it in EX → 1 stall (load-use)
$I_2 \to I_4$ forwarded, no stall. $I_4 \to I_5$ ALU to ALU

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forwarded, no stall. Cycles = 12 + 1 = $\boxed{13}$

(i.v) Delay slot schedule:

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Loop:  LD    R4, 0(R3)
       LD    R5, 4(R3)
       ADDI  R2, R2, #1
       ADD   R6, R4, R5
       SUBI  R3, R3, #8
       BNEZ  R2, Loop
       MUL   R4, R6, R7  (delay slot)
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(ADD R_{11}, R_{12}, R_{13} after the loop)

ADDI moved b/w the loads & the ADD, so no load-use stall. BNEZ only needed R2, ADDI is 3 inst. before it, forwarded, not stall. MUL doesn't affect the branch, so it fills the delay slot & still runs every iteration. SUBI moved before the branch. Result: 7 cycles per iteration. 0 stalls, delay slot not wasted.

(Q4) Branches 17%, mispredicted = 100 - 41 = 59%

branch stalls = 0.17 × 0.59 × 1.7 = 0.1705

other stalls = 0.02 × 1.3 = 0.026 = $\boxed{1.1965}$

(Q5)(i) RAW: $I_1 \rightarrow I_2$ (F_2)
$I_2 \rightarrow I_3$ (F8)
$I_3 \rightarrow I_5$ (F_{12})
$I_5 \rightarrow I_6$ (F20)
$I_6 \rightarrow I_7$ (F24)
$I_7 \rightarrow I_9$ (F28)
$I_9 \rightarrow I_{10}$ (F32)

WAR: $I_2 \rightarrow I_4$ (F2)
$I_2 \rightarrow I_4$ (R2)

WAW: $I_1 \rightarrow I_4$ (F2)
$I_1 \rightarrow I_{11}$ (F2)
$I_4 \rightarrow I_{11}$ (F2)

Structural:-

- $I_1, I_3, I_7, I_{10}, I_{11}$ share the one FP adder. $I_{2,4,9}$ share the one FP multiplier. Both pipelined (II = 1), conflict only if ~~the~~ two start in the same cycle.
- divider not pipelined (II = 25): I_5 & I_6 cannot overlap.
- One reg. write port: out of order completion (add 3, mul 6, div 24) can put two inst. in WB in the same cycle.

Control:-

I_8 resolved in integer EX, so I_9 & I_{10} are fetched before the outcome is known (control dependent on I_8). If taken they are wrong path → 2 cycle penalty, next is Target I_{11}.

(ii) Solution:

RAW: forwarding (EX output → next EX input) + interlock stalls for the FP latency (add 3, mul 6, div 24).

WAR: no hazard, in-order issue & operands read in ID.

WAW: no hazard here, I_1 finishes before I_4 & I_4 before I_{11} (interlock would stall otherwise).

Structural:-

I_6 stalls until the divider is free (EX start

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18 + 25 = 43). Checked all WB cycles, no two in the same cycle~~s~~.

Control:-

predict not taken. Branch resolved in EX at cycle 69. If taken: flush I_9 & 10 (2 cycles) & fetch I_{11} at ~~E~~ 70 (I_{11} WB = 77).

A = FADD EX (4 cycles)
M = FMUL EX (7 cycles).
D = FDIV EX (25 cycles).

I_1	
I_2	
I_3	
I_4	
I_5	
I_6	
I_7	
I_8	
I_9	
I_{10}	
I_{11}	

CC	1	2	3		7	8	9	10		17	18	19	20	21			29		48	49			75	76	77	78		82	83	84	8
I1	IF	ID	Ex	3-6	M	WB																									
I2		IF	stall..				ID	Ex	10-16	M	WB				24	25															
I3							IF	Stall...				ID	EX	20-23	M	WB															
I4												IF	ID	EX	21-27	28 M	29 WB														
I5													IF	ID	Ex	12-46		47 M	48 WB												
I6														IF	stall...					ID	Ex	50-74	M	WB							
I7																				IF	stall				ID	EX	78-80	M	WB		
I8																									IF	stall			ID	Ex	
I9																													IF	stall	
I10																															
I11																															I

	90		97	98	99			104				
89												
ID	Ex	92-96	M	WB								
IF	stall			↳	ID	Ex	100-103	M	WB			
IF^{86}	Ex$^{87-90}$		M^{91}	WB^{93}								

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